

components and double-check for any overlooked error.

Operation. The functions of all keys are given in Table I. The mode selection switch S7 selects either repeat or single mode. Single mode allows user to run the timer operation in 'on' time and 'off' time sequence once. In repeat mode of operation, the timer repeats 'on' time and 'off' time sequence continuously. During this cycle if this switch is changed to single-mode, the timer stops as the cycle completes. Also, if the emergency stop (ES) button is pressed during any mode of operation the timer operation will stop.

The step-by-step operation when the main circuit is powered is as follows:

1. The 'enter on time' message is displayed on LCD
2. User has to enter the desired time by incrementing/decrementing time using 'Inc.' (S2) / 'Dec.' (S3) keys
3. After pressing the 'Enter' (S4) key the user will be prompted to enter 'off' time
4. Using the same S2 and S3 keys, the 'off' time may be entered and the 'Enter' key pressed

TABLE I
Functions of Push Buttons

Switch number	Switch name	Function
S1	Start	Starts timer operation
S2	Inc. time	Increments time set by 1 sec. max limit is 60
S3	Dec. time	Decrements time set by 1 sec. min limit is 1
S4	Enter	Used to enter time set value
S5	ES (RST)	Emergency stop or system reset.
S6	Control selection	Selection of either remote or keypad control
S7	Mode selection	Selection of either repeat or single control

TABLE II
Status of Data Input Pins and the Codes Transmitted

Switch	D3	D2	D1	D0	Code
start	0	1	1	1	7
Inc. time	1	0	1	1	B
Dec. time	1	1	0	1	D
enter	1	1	1	0	E
ES	1	1	1	1	F

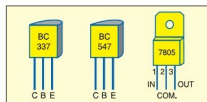


Fig. 4: Pin configurations of BC337, BC547 and 7805

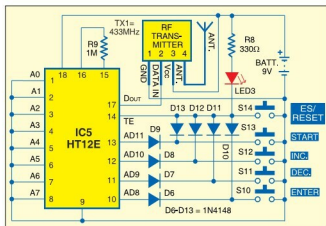


Fig. 5: Circuit diagram of RF transmitter

5. 'Press Start' message is displayed as the user enters the time
6. After pressing 'Start' (S1) key the operation starts
7. The relay is energised and the device remains on till the 'on' time counts down to 0. After that the relay is de-energised and the device turns off. It remains in this state till 'off' time counts down to 0

8. If the timer is operating in the repeat mode the cycle will repeat continuously and device will be switched on and off after required time intervals. In this mode if the operation has to be stopped then ei-

ther switch S7 has to be toggled or ES button has to be pressed

9. If the timer is operating in single mode then as one on-off time cycle completes, the timer stops working. One has to enter 'on' time and 'off' time again to re-start operation

10. For remote control operation,

port pins P1.0 through P1.3 of microcontroller AT89C51 are changed to D11 through D8 of HT12D, respectively, using four-pole double-throw (4PDT) switch S8. Also, switch S6 connected to port pin P3.5 is changed to remote position to set remote operation mode.

Remote control transmitter and receiver circuit. Remote control transmitter and receiver are made using readily available encoder (HT12E) and decoder (HT12D) chips.

Transmitter circuit. Fig. 5 shows the circuit diagram of transmitter consisting of two main components—encoder HT12E (IC5) and radio frequency transmitter module 433MHz (TX1). All the address pins A0 through A7 are tied to ground to set address '00h.' A one mega-ohm resistor is connected between oscillator pins 15 and 16 of HT12E. D_{OUT} pin17 of HT12E is connected with input of ASK transmitter (TX1). Switches S10 through S13 are connected with data input pins AD8 through AD11 of HT12E in such a way that when you press any key, TE pin is automatically grounded. Key S14 is directly connected between TE pin and ground as shown Fig. 5. LED3 is connected for indication of pressing the switch. An actual-size, single-side PCB for RF transmitter is shown in Fig. 6 and its component layout in Fig. 7.

Operation. When you set the address and data inputs on HT12E and